

FORECASTING INDIA'S INFLATION

Objective

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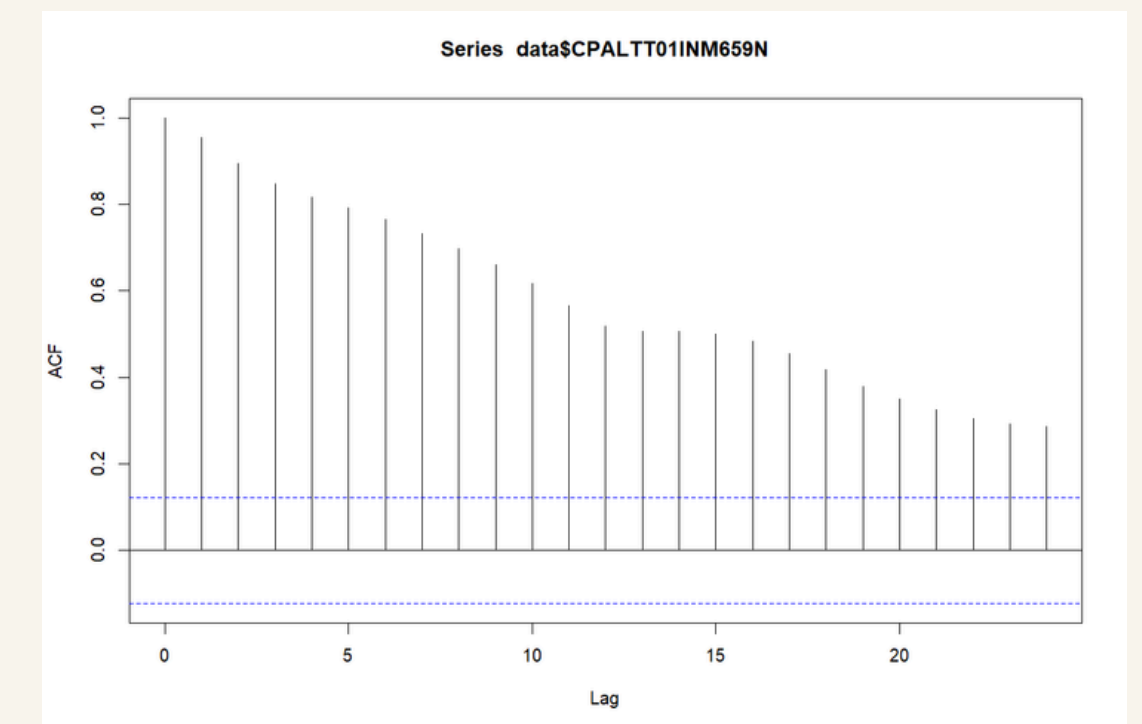
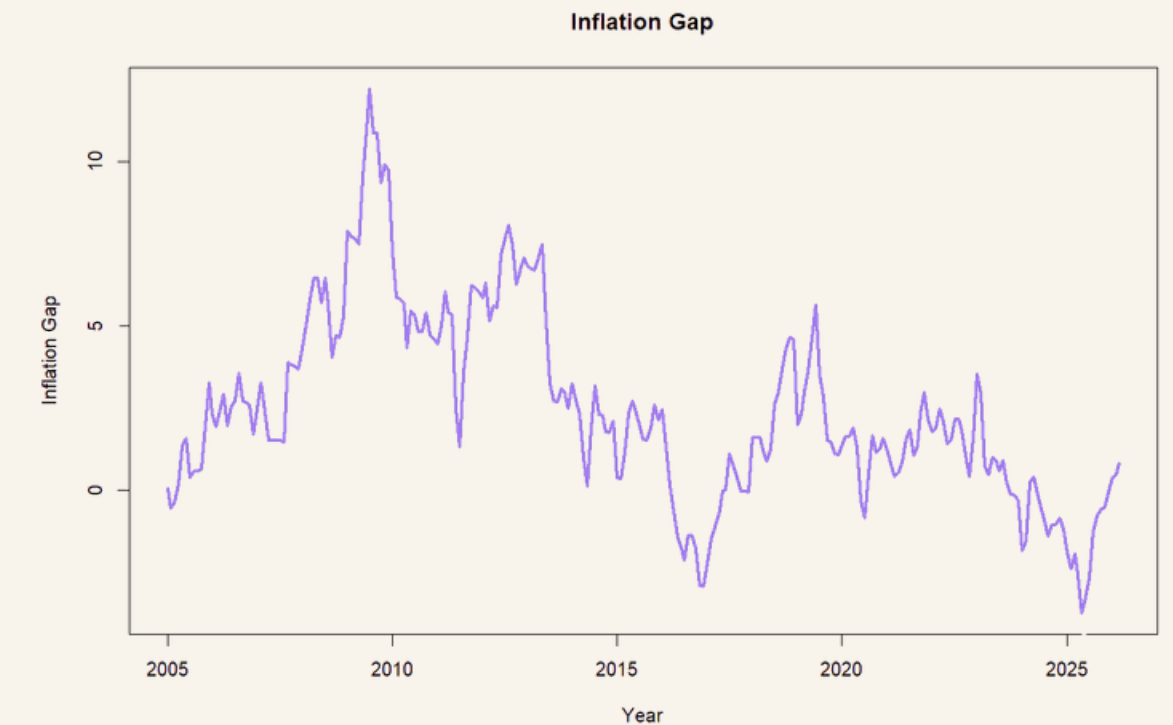
- Forecast India's inflation gap for September, October and November 2026
- Use monthly inflation data and a time-series model to generate the forecasts

Data

- Monthly Indian inflation data from 07-2005 to 08-2026 (FRED+MOSPI)
- 255 observations
- Inflation gap calculated by subtracting the 4% target from inflation

Preparing the Data

- Converted the date column from Excel serial numbers into dxates readable by R
- Converted the inflation column from character format into numeric format
- Removed observations containing missing values



The graph does not show overt signs of seasonality

Testing for Stationarity

Augmented Dickey-Fuller (ADF) Test		
Statistic	Lag Order	p-value
-3.1626	6	0.09457

ADF

Null hypothesis: The series has a unit root → non-stationary

- ADF p-value = 0.0946
- At the 5% level, I fail to reject the null
- Therefore, ADF does not provide sufficient evidence of stationarity

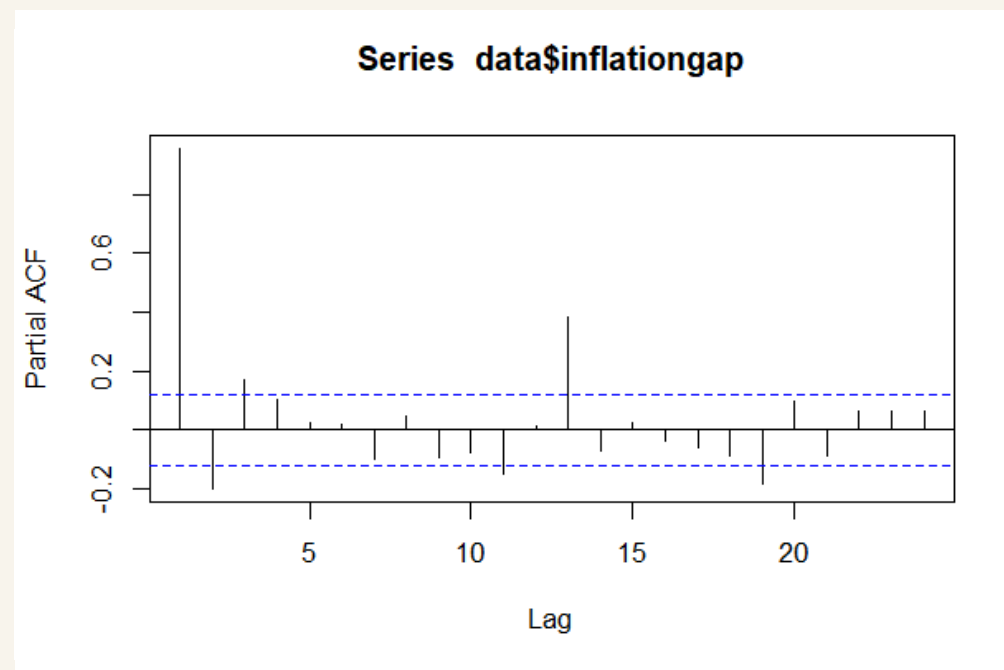
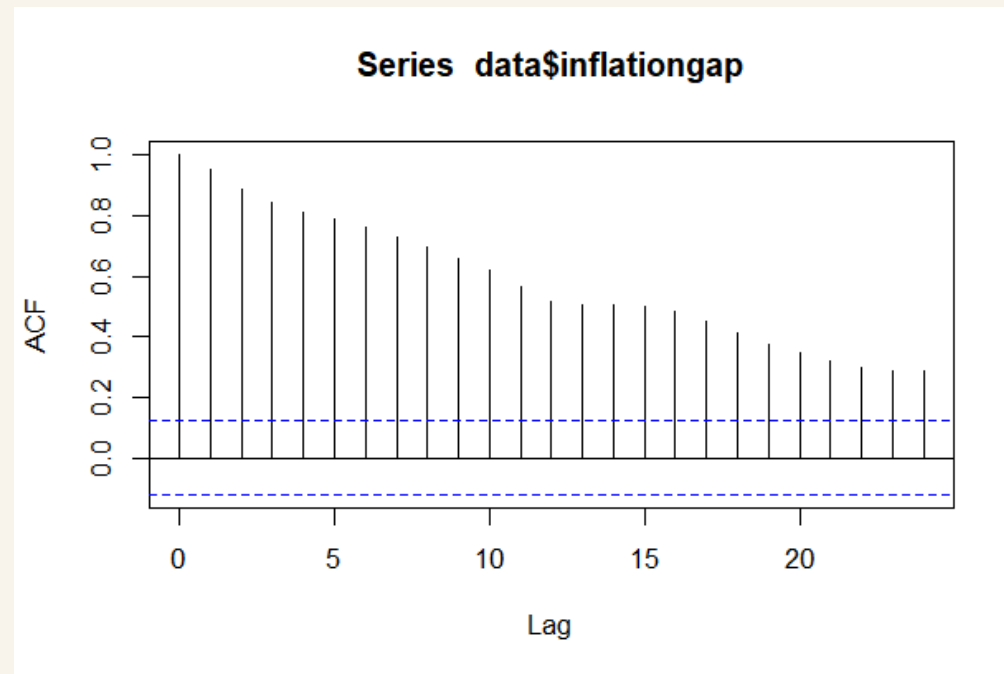
KPSS Test		
Test Statistic	Truncation Lag	p-value
1.7203	5	< 0.01

KPSS Test

Null hypothesis: The series is stationary

- KPSS p-value = 0.01
- I reject the null
- Therefore, KPSS provides evidence against stationarity

ACF/PACF



ACF is useful for identifying possible MA behaviour
 PACF is useful for identifying possible AR behaviour

- The ACF showed a gradual decline rather than a clear cutoff
- The PACF also did not give a clear ARMA specification
- So we proceed with EACF

AR/MA

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1	x	x	x	o	o	o	o	o	o	x	o	x	x	o
2	x	x	x	o	o	o	o	o	o	x	o	x	x	o
3	x	x	o	o	o	o	o	o	o	o	o	x	x	o
4	x	x	o	o	o	o	o	o	o	o	o	x	x	o
5	x	o	x	o	o	o	o	o	o	o	o	x	x	x
6	x	x	x	o	o	x	o	o	o	o	o	x	x	o
7	x	x	x	o	o	o	o	o	o	o	o	x	x	o

- EACF helps identify possible ARMA(p,q) combinations.
- Chosen models:
 - ARMA(1,3)
 - ARMA(2,3)
 - ARMA(3,2)

Model	AIC	BIC
ARMA(2,3)	626.33	653.0923
ARMA(3,2)	623.9	650.666
ARMA(1,3)	630.01	653.2367

Interpretation

- Lower AIC/BIC indicates a better balance between model fit and complexity
- ARMA(2,3) had the lowest AIC and BIC

Residual Diagnostic

- For ARMA(2,3):
- Ljung–Box p-value = 0.3412
- Since $p > 0.05$, I fail to reject the null hypothesis of white-noise residuals

Estimating
Models

Does the Model Actually Forecast Well?

A model can fit the existing data well but still produce poor forecasts.
So I tested the models using a 12-month holdout sample.

Data split

- Total observations = 255
- Training observations = 243
- Holdout observations = 12

I estimated each model using the first 243 observations and then forecast the final 12 observations.

Model	MAE	RMSE	Best RMSE
ARMA(1,3)	0.4567	0.6461	No
ARMA(2,3)	0.4193	0.5975	Yes
ARMA(3,2)	0.5771	0.646	No

Important comparison
Choosing **ARMA (2,3)**

Forecasting September–November

Month	Forecast (%)	Lower 95%	Upper 95%
9/26/2026	4.955	3.374	6.536
10/26/2026	5.012	2.601	7.423
11/26/2026	5.073	2.18	7.967

